

Tropical Cyclone: Steady State Physics

Kerry Emanuel's Potential Intensity Theory

MEA 458/NCSU

Learning Outcomes

Be able to:

- ▶ Recall the angular momentum structure of a mature hurricane
- ▶ Describe the Carnot heat engine and the maximum that is possible
- ▶ Describe the maximum intensity of a hurricane based on the Carnot engine model for it
- ▶ Identify the major physical assumptions and weaknesses of PI theory.

Consider a parcel of air that is swirling around the center of a hurricane. The angular momentum of the parcel of air in the vertical direction is composed of two parts: (a) Relative angular momentum owing to the circular motion around the center of the storm; and (b) the planetary angular momentum. At any distance (radius) from the hurricane center, we write the Absolute angular momentum to be:

$$M = \frac{1}{2}fr^2 + Vr \quad (1)$$

Where $f = 2\Omega\sin\phi$, is the Coriolis parameter (or better still, the vertical component of planetary vorticity) and V is the tangential speed of the air parcel.

If a parcel swirling towards the center within the boundary layer (the lowest layer of the atmosphere where it feels the effect of surface friction) were to conserve its angular momentum, then we can estimate the tangential speed at any radius using the simple relation

$$M_1 = M_2 \quad (2)$$

If we assume that the parcel starts with $V_1 = 0$ at some radius r_1 , then its tangential speed at some other radius

Thought experiment: Assume that a parcel starts out at 1000km from the center and reaches the eyewall at around 10 km, then angular momentum conservation would yield (assuming $f = 1.e^{-5}$)

$$V_2 \approx \frac{1}{2} 10^{-5} \frac{10^{12}}{10^4} \approx 500 \text{m/s} \quad (5)$$

Of course, we do not see such high wind speeds in the eye. This is because parcels lose angular momentum due to friction in the lowest layer. The angular momentum is lost to the ocean below.

Absolute Angular Momentum Structure of a hurricane

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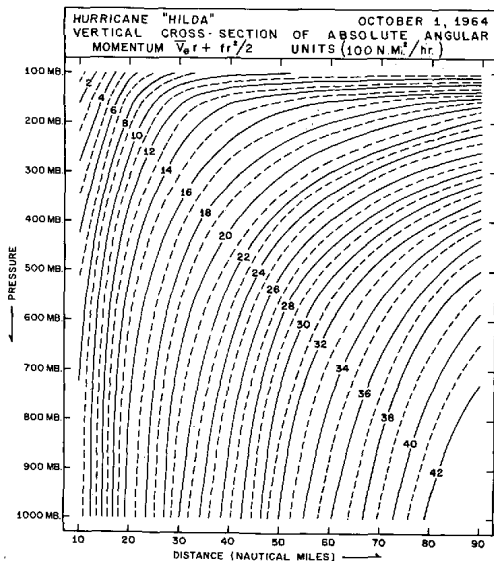


Figure: Source: Hawkins (1968), Mon. Wea. Rev.

Absolute Angular Momentum Structure of a hurricane

We find that the absolute angular momentum of a hurricane has a very interesting structure. It decreases inward. The lines (surfaces really in 3-D), flare outward.

Central idea

Potential intensity is an estimate of the maximum intensity that a mature tropical cyclone could maintain in a given thermodynamic environment.

PI is useful as

- ▶ a thermodynamic speed limit,
- ▶ a climate-scale intensity constraint,
- ▶ a normalization for observed storm intensity,
- ▶ a diagnostic of environmental favorability.

PI is not, by itself,

- ▶ a genesis theory,
- ▶ an intensification-rate theory,
- ▶ a deterministic forecast of one storm,
- ▶ a complete theory of inner-core structure.

Foundational sources: Emanuel (1986, 1995); Bister and Emanuel (1998, 2002).

Idealized energy cycle in a hurricane

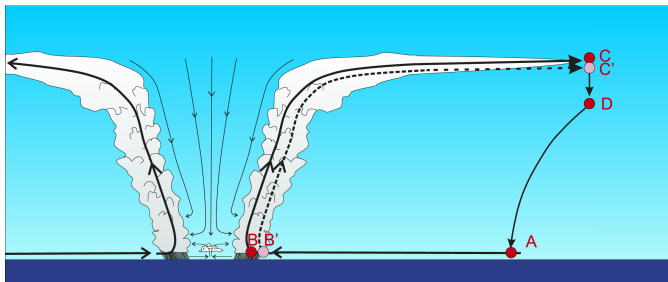


Figure: Source: Emanuel (2018)

Idealized energy cycle in a hurricane

- ▶ Air spirals in at low-levels (A to B) at early constant temperature. Now, this is a very important stage. The air is flowing towards the center, towards lower pressure. So it will tend to cool adiabatically. However, heat (enthalpy) input from the ocean nearly compensates the cooling.
- ▶ Air rises from B to C. It conserves its angular momentum as well as its θ_e . If we also assume that the air is saturated, then its rise will be moist adiabatic.
- ▶ Air subsides from c to D nearly isothermally and then D to A adiabatically.

A key point in the life-cycle presented in the previous slide is that the primary energy input comes from the interaction of air and the ocean surface below.

How is this energy transferred from the ocean to the air?

In the simplest sense, the gusty winds lead to evaporation of water (thus carrying all the latent heat into the air) and also sensible heat transfer from the warm ocean below. Additionally, frictional energy dissipation (i.e. loss of energy due to friction) is, in part, converted to thermal energy that is taken in by the air.

How much enthalpy is transferred from the ocean to the air?

The enthalpy of air (per unit mass) can be written as:

$$k = c_p T + L_v q \quad (6)$$

Where c_p is the heat capacity at constant pressure and L_v is the latent heat of vaporization.

How much enthalpy is transferred from the ocean to the air?

A rough estimate of the enthalpy transferred to air is given by this formula:

$$\rho C_k |V| [k_o^* - k] \quad (7)$$

Here $|V|$ is the wind speed. ρ is air density, k is the enthalpy of air, and k_o^* is the saturation enthalpy at the sea surface, and can be thought of a thin layer of air that is touching the sea surface. The difference $[k_o^* - k]$ represents the thermal disequilibrium between the ocean and the air above it.

C_k represents a quantity called enthalpy exchange coefficient (more on this later).

Hurricane as a Carnot Engine

The preceding picture of a hurricane lifecycle resembles a Carnot engine

Hurricane as a Carnot Engine

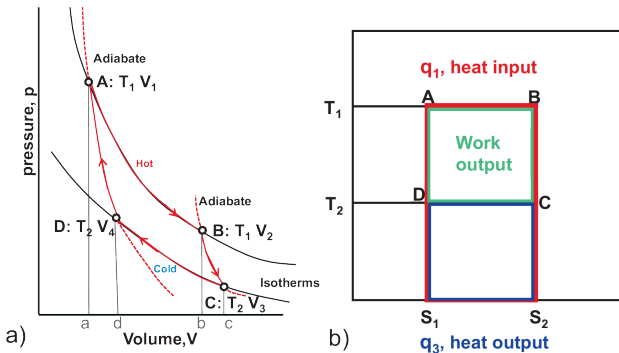


Figure: Source:

<https://www.tf.uni-kiel.de/matwis/amat/td_kin_i/kap₁/backbone/rs_e31.html>

In a a Carnot engine: Heat input occurs during the isothermal expansion stage: This is the leg A to B of a hurricane shown in earlier slide.

Hurricane as a Carnot Engine

For a Carnot Engine operating between the temperature limits T_1 (warm) and T_2 (cold), the maximum work that can be extracted is given by:

$$W = Q \frac{T_1 - T_2}{T_1} = \epsilon Q \quad (8)$$

Where Q is the heat input and W is the mechanical energy and

$$\epsilon = \frac{T_1 - T_2}{T_1} \quad (9)$$

Hurricane as a Carnot Engine

In a hurricane, we identify two sources of heat input:

- ▶ Enthalpy transfer from the ocean: $\rho C_k |V| (k_o^* - k)$
- ▶ Heat generated due to friction (dissipative heating). This scales as the square of the wind
 $C_D |V|^3$
 C_D represents a quantity called drag coefficient (more on this later)

Hurricane as a Carnot Engine

For a hurricane, assuming all the energy transfer occurs at the eye-wall, where the winds are fastest $V = V_m$, we can write

$$W = \epsilon[\rho C_k |V_m| (k_o^* - k) + C_D |V_m|^3] \quad (10)$$

Assuming steady state, we can claim that all work done by the engine is balanced by the dissipative heating - that is friction balances the work done. This is somewhat similar to a rain drop falling at terminal velocity. It falls at a steady speed when the downward gravity is exactly balanced by the drag felt by it.

So under steady state, we say:

$$W = C_D |V_m|^3 \quad (11)$$

So, from equations 10 and 16, get:

$$C_D |V_m|^3 = \epsilon [\rho C_k |V_m| (k_o^* - k) + C_D |V_m|^3] \quad (12)$$

Solving for the maximum wind speed, we get:

$$|V_m|^2 = \frac{\epsilon}{1 - \epsilon} \frac{C_k}{C_D} (k_o^* - k) \quad (13)$$

Hurricane as a Carnot Engine

One last thing, what is ϵ for the hurricane. Since it depends on the temperature limits of the Carnot cycle, we can say:

$$\epsilon = \frac{T_s - T_o}{T_s} \quad (14)$$

Where T_s is the Sea Surface temperature and T_o is the outflow temperature. And:

$$\frac{\epsilon}{1 - \epsilon} = \frac{T_s - T_o}{T_o} \quad (15)$$

Finally the peak winds in the hurricane can be written as:

$$|V_m|^2 = \frac{T_s - T_o}{T_o} \frac{C_k}{C_D} (k_o^* - k) \quad (16)$$

This is a pretty amazing estimate of the hurricane peak winds as a function of minimal number of external parameters. Obviously, it is great estimate, but can't predict the exact value of V for a given storm. Why?

Note: In practical calculation, a more realistic approximation involves the CAPE calculated from the full atmospheric profile

Numerical example

Suppose

$$T_s = 300 \text{ K}, \quad T_o = 200 \text{ K}, \quad k_s^* - k = 20,000 \text{ J kg}^{-1},$$

with $C_k/C_D = 1$.

Then

$$V_p^2 = 1 \times \frac{300 - 200}{200} \times 20,000 = 10,000 \text{ m}^2 \text{ s}^{-2},$$

so

$$V_p \approx 100 \text{ m s}^{-1}.$$

Interpretation

This is an idealized thermodynamic upper bound. It is not a forecast that a real storm will reach 100 m s^{-1} .

Observed intensity versus PI

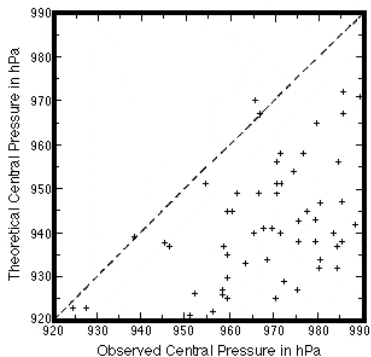


Figure: Observed minimum central pressures (abscissa) versus potential minimum pressure (ordinate) calculated using climatological SSTs and atmospheric profiles at the locations of observed Atlantic hurricanes. Courtesy of Dr. Lars Schade. Taken from Kerry Emanuel's webpage: <https://emanuel.mit.edu/hurricane-potential-intensity-maps-0/>

A diagnostic is the fraction of PI realized by a storm:

$$RI = \frac{V_{\max}}{V_p}.$$

- ▶ $RI \ll 1$: storm is well below its thermodynamic ceiling.
- ▶ $RI \sim 1$: storm is close to the PI envelope.
- ▶ Apparent $RI > 1$ can occur because of local disequilibrium, storm motion, observational uncertainty, boundary-layer supergradient winds, or missing physics in the PI calculation.

Useful interpretation

PI constrains the intensity distribution, but individual storms often fail to reach PI because of shear, ventilation, ocean feedback, internal dynamics, and finite intensification time.

Major limitation: PI is an equilibrium theory

PI assumes that the storm can approach a mature, steady state in a fixed environment.

Real storms usually experience

- ▶ translation over changing SST and ocean heat content,
- ▶ changes in vertical shear and humidity,
- ▶ eyewall replacement cycles,
- ▶ land interaction,
- ▶ finite time for intensification.

A storm can temporarily satisfy

$$V_{\max} > V_p(\text{local environment})$$

without violating the spirit of PI, because the local environment may have changed faster than the storm could adjust.

Major limitation: shear and ventilation are not in the basic formula

Classical PI contains no explicit vertical wind shear term:

$$|\mathbf{V}_{200} - \mathbf{V}_{850}|.$$

But shear can reduce intensity by importing low-entropy environmental air into the inner core.

Ventilation view

Dry or low-entropy air entering the core lowers eyewall entropy and reduces the fraction of PI realized by the storm.

Conceptually,

actual intensity $\approx$ PI ceiling—ventilation penalty—ocean penalty—internal-dynam

Major limitation: the ocean is not fixed

Standard atmospheric PI treats SST as prescribed, but a hurricane changes the ocean beneath it.

wind stress $\rightarrow$ mixed-layer deepening $\rightarrow$ cold-water entrainment $\rightarrow \Delta SST < 0$.

This reduces

$$k_s^* - k$$

and therefore reduces

$$V_p^2 \propto (k_s^* - k).$$

The strength of the feedback depends on translation speed, storm size, mixed-layer depth, stratification, and ocean heat content.